ITPM

IT Project Management

Unit 2 — Project Audit, Risk, Resource & Cost Control

Simple, Easy-to-Understand Notes

1. Project Audit

A project audit is a careful examination of a project to check its purpose, its effect (output), and whether it is meeting its objectives.

Objectives of Project Audit

- Check whether the objectives (purpose) of the project are being achieved
- Evaluate cost, schedule, and quality of the project
- Identify problems and risks in the project
- Cross-check the necessary actions taken
- Evaluate overall project performance

1.1 Areas Covered in Project Audit

#	Area	What is Checked
1	Project Planning	Check if the project is realistic; review the budget, schedule, and timeline
2	Project Execution	Examine actual progress vs planned progress
3	Cost and Budget	Check if resources are used properly; compare actual expenditure vs approved budget
4	Quality	Check whether the project meets quality standards
5	Risk Management	Check if risks were identified and the right action was taken to resolve them
6	Team Performance	Evaluate team members — their roles and coordination

2. Project Termination

Project termination is the process of ending a project — either because it is completed, or because it is no longer beneficial to continue.

Types of Project Termination

Type	Meaning
1. Normal Termination	The project is completed successfully and its objective is fully achieved (the natural end).
2. Premature Termination	The project ends before it is fully complete because there is no budget, resources, or the requirements have changed.
3. Perpetual Termination	The project continues indefinitely without reaching any complete/finishing point (it just keeps running).
4. Failed Termination	The project ends without fulfilling its objective — because it became too expensive, technically impossible, or is no longer useful.

Steps of Project Termination

1. Decide to terminate the project
2. Review project status
3. Complete documentation
4. Release project resources

5. Close contracts and financial accounts
6. Transfer the project to the customer
7. Prepare the final report
8. Conduct a post-project review
9. Document lessons learned
10. Formally close the project

3. Resource Scheduling

Resource scheduling means allocating available resources to different tasks/activities of a project, keeping in mind time, cost, and quality — so that the project is completed properly and achieves good quality within budget.

Types of Resources

#	Resource Type	Examples
1	Human Resources	Developers, testers, project managers, designers, analysts
2	Hardware (H/W) Resources	Network devices, servers
3	Software (S/W) Resources	Databases, testing tools
4	Financial Resources	Project budget and expenses
5	Time	Working hours, project duration

Steps in Resource Scheduling

11. Identify the activities — divide the project into parts (Part 1, Part 2, Part 3, Part 4...)
12. Identify resources needed for each task — HR, H/W, S/W, Cost
13. Estimate resources — how much resource and time is required
14. Allocate resources to the tasks
15. Prepare the resource schedule
16. Monitor the usage of resources

4. Resource Leveling

Resource leveling is the concept of adjusting the flow (schedule) of a software project as per the availability of resources, so that no resource is overloaded.

Objectives

17. Estimate resource over-allocation
18. Balance the workload
19. Improve resource utilization

Example: Task A (Develop) is scheduled Monday–Wednesday and Task B (Develop) is scheduled Tuesday–Thursday, using the same developer. Since the same resource cannot do both tasks at the same time, resource leveling adjusts (shifts) the schedule to remove this overlap.

Advantages

- Reduces resource overloading
- Improves productivity
- Reduces stress in the work environment

5. Project Cost Control

Project cost control is the process of monitoring and controlling project expenses so that the project stays within the approved budget and time.

5.1 PERT / Cost Technique

PERT/Cost is a technique used to find the estimated cost of a project, using three cost estimates (similar to PERT time estimates):

Cost Estimate	Meaning
Optimistic Cost (Co)	The lowest possible cost
Most Likely Cost (Cm)	The most probable / expected cost
Pessimistic Cost (Cp)	The highest possible cost

Main Activities in Project Cost Control

20. Cost Estimation — predicting the cost required
21. Budget Preparation — preparing the approved budget
22. Cost Monitoring — tracking actual cost vs approved budget regularly
23. Corrective Action — taking action if cost is going over budget

6. Risk Analysis & Risk Management

Risk is any uncertain event or condition that, if it happens, can have a negative (or sometimes positive) effect on the project's objectives — such as its cost, schedule, or quality.

Risk Management is the process of identifying risks and taking the right action to resolve or reduce them, so the project is not badly affected.

6.1 Risk Management Process

Step	Name	Meaning
1	Risk Identification	Finding out what risks could affect the project (e.g., budget shortage, technical failure, delay in schedule)
2	Risk Analysis	Studying each risk to understand how likely it is to happen and how much damage it can cause
3	Risk Prioritization	Ranking risks — from the most serious/likely to the least serious
4	Risk Response Planning	Deciding what action to take for each risk (avoid, reduce, transfer, or accept it)
5	Risk Monitoring & Control	Keep watching the risks throughout the project and update the plan if needed

6.2 Types of Risk Analysis

Type	Meaning
Qualitative Risk Analysis	Ranks risks based on their probability (chance of happening) and impact (how serious the effect is) — usually described in words like High, Medium, Low

Quantitative Risk Analysis	Uses numbers/data to calculate the actual probability and cost impact of a risk

6.3 Common Ways to Respond to Risk

Response	Meaning
Avoid	Change the plan so the risk does not happen at all
Reduce (Mitigate)	Take action to lower the chance or effect of the risk
Transfer	Shift the risk to someone else (e.g., insurance, outsourcing)
Accept	Take no action, but keep monitoring — used for small/low-impact risks

Simple way to remember Risk Management: Identify → Analyze → Prioritize → Plan Response → Monitor.

Quick Revision Summary

Topic	One-line Takeaway
Project Audit	Examines project purpose, output, and objectives — covers planning, execution, cost/budget, quality, risk, and team performance
Project Termination	4 types: Normal, Premature, Perpetual, Failed — ends with 10 formal closing steps
Resource Scheduling	Allocating resources (Human, H/W, S/W, Financial, Time) to project tasks in 6 steps
Resource Leveling	Adjusts the project schedule as per resource availability to avoid overloading
Project Cost Control	Monitors project expenses using PERT/Cost (Optimistic, Most Likely, Pessimistic cost)
Risk Analysis / Management	Identify → Analyze → Prioritize → Plan Response → Monitor; responses = Avoid, Reduce, Transfer, Accept